Charge-and-Release Rhythms in Ore-Forming Systems: A Random Matrix Theory Test of Metallogenic Timing across Four Provinces, Including a Non-Magmatic Control

Single Ore Systems Show Level Repulsion Whether or Not Magmatism Is Involved

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Abstract

We test whether the timing of ore-forming pulses within hydrothermal mineral deposits carries the level-repulsion signature of random matrix theory (RMT), extending a program that found GOE-class repulsion in geological boundaries, seismotectonic rhythms, and mantle-plume eruptions. The motivation is a charge-and-release hypothesis: each mineralization pulse depletes the local metal, fluid, and heat budget of an ore system, which must re-accumulate before the next pulse, imposing a minimum recurrence interval analogous to eigenvalue repulsion. Using compiled multi-stage geochronology from *four* metallogenic provinces—Andean porphyry copper (subduction arc; 11 districts), South China Nanling W–Sn (intracontinental granite; 15 deposits), Tethyan porphyry copper (collisional; 9 districts), and orogenic gold (metamorphic, *non-magmatic*; 10 deposits across the Yilgarn, Abitibi, and West African cratons)—we contrast single-source and mixed configurations. **Single sources show level repulsion in all four provinces**: pooled spacing ratios are $\langle r \rangle = 0.60$ (Chile), 0.57 (Nanling), 0.71 (Tethyan), and 0.68 (gold), with bootstrap 95% confidence intervals that all exclude Poisson (0.386). **Mixed** configurations collapse toward Poisson ($\langle r \rangle = 0.31$–0.47). The source-separation effect is consistent across all five deep-Earth systems we have now studied (mantle plumes plus these four). Crucially, *orogenic gold—whose fluids derive from metamorphic devolatilization, not magmatism—shows the same repulsion*, which cuts off the explanation that the signal is merely inherited from magmatic-intrusion cyclicity and supports an intrinsic charge-release memory. We argue for a unifying principle: GOE-class repulsion emerges wherever a single long-memory process is isolated, and Poisson emerges wherever many independent sources superpose. We remain explicit that the compiled ages are representative literature values, not a systematic census, and that GOE-versus-GUE is not resolved at present sample sizes.

Keywords: random matrix theory; metallogeny; porphyry copper; tungsten-tin; orogenic gold; ore geochronology; level repulsion; spectral superposition

Code/data: https://github.com/Ruqing1963/metallogeny-rmt
Companion studies: https://zenodo.org/records/20766310 ; https://zenodo.org/records/20768130 ; https://zenodo.org/records/20768420

1 Introduction

Ore deposits form in pulses. A single porphyry copper system, granite-related W–Sn deposit, or orogenic gold camp typically records several discrete mineralization stages (Sillitoe, 2010; Chelle-Michou et al., 2017). Whether the timing of these pulses is random or carries memory is a question about the dynamics of ore-forming systems.

We propose a *charge-and-release hypothesis*: each major mineralization pulse depletes the local reservoir of metal, fluid, and heat, which must then re-accumulate before the next pulse can occur. This imposes a minimum recurrence interval and a suppression of short inter-pulse spacings—the level repulsion of random matrix theory (Wigner, 1957; Mehta, 2004). The mechanism is structurally identical to the thermal-shadow mechanism we proposed for mantle plumes (Chen, 2026c): a pulse exhausts a local resource that must be replenished.

This study extends a three-racetrack RMT program (Chen, 2026a,b,c) to metallogeny. A recurring lesson is the spectral superposition theorem: the sum of many independent renewal processes tends to Poisson (Daley & Vere-Jones, 2003), so single-source signals are erased by mixing. We design around source isolation.

The central scientific subtlety of metallogeny is that ore formation in magmatic-hydrothermal systems is externally triggered by tectonic-magmatic events (Sillitoe, 2010), so a GOE signal could be *inherited* from magmatic cyclicity rather than reflecting intrinsic charge-release memory. We address this directly by including a non-magmatic control: orogenic gold deposits, whose ore fluids derive from metamorphic devolatilization during orogenesis, with no direct genetic link to magmatism (Goldfarb et al., 2005). If orogenic gold also shows repulsion, the signal cannot be merely inherited from magmatic intrusion timing.

2 Data and Methods

2.1 Datasets

We compiled multi-stage absolute ages for four provinces. (1) *Andean porphyry copper*: 51 ages, 11 districts, dominated by Re–Os molybdenite (Maksaev et al., 2004; Romero et al., 2011; Deckart et al., 2013). (2) *South China Nanling W–Sn*: 51 ages, 15 deposits, combining zircon U–Pb, molybdenite Re–Os, cassiterite U–Pb, and muscovite Ar–Ar (Mao et al., 2017; Zhou et al., 2016). (3) *Tethyan porphyry copper*: 30 ages, 9 districts spanning Tibet (Gangdese/Yulong), Iran (Kerman/Arasbaran), and Pakistan (Chagai) (Hou et al., 2009; Yang et al., 2017). (4) *Orogenic gold*: 32 ages, 10 deposits across the Yilgarn (Australia), Abitibi (Canada), and West African cratons, using U–Pb xenotime/monazite, Re–Os sulfide, and Ar–Ar (Vielreicher et al., 2010; Pelletier et al., 2022; Oberthür et al., 1998). Each district/deposit is treated as a single ore-forming source.

Data caveat. These are representative ages from the published literature and reviews, not a systematic census, and suffice for a feasibility-level test.

2.2 RMT analysis

We sort ages, compute inter-event intervals, normalize each source to unit mean (unfolding), pool, and evaluate the spacing ratio $\langle r \rangle$ (Atás et al., 2013) (Poisson 0.386, GOE 0.536, GUE 0.603), KS tests, and a bootstrap 95% CI. Per-source unfolding prevents differing mean rates from inducing structure.

3 Results

3.1 All four provinces show single-source repulsion

Pooling per-source unfolded intervals yields level repulsion in every province (Table 1, Figure 1). All four bootstrap CIs exclude Poisson; KS tests reject Poisson and prefer GOE/GUE.

Table 1: Single-source (pooled) versus mixed configurations across four metallogenic provinces. Every province shows repulsion when sources are isolated and collapses when mixed.

Province (setting)	Config.	n	$\langle r \rangle$	$\hat{\beta}$	95% CI	Mixed $\langle r \rangle$
Andean porphyry Cu (subduction)	Single	40	0.601	1.97	[0.471, 0.670]	0.470
Nanling W–Sn (intracontinental)	Single	36	0.574	1.57	[0.488, 0.685]	0.307
Tethyan porphyry Cu (collision)	Single	21	0.712	3.00	[0.510, 0.797]	0.391
Orogenic gold (*non-magmatic*)	Single	22	0.678	2.75	[0.606, 0.823]	0.432

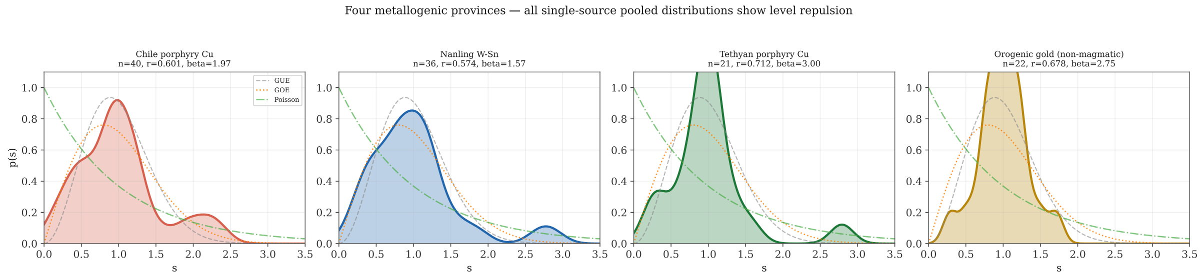

Figure 1: Single-source pooled spacing distributions for all four metallogenic provinces. Each lies in the GOE–GUE range; none is Poisson.

3.2 The non-magmatic control

Orogenic gold, whose fluids derive from metamorphic devolatilization with no direct magmatic link, shows repulsion as strong as the magmatic provinces ($\langle r \rangle = 0.678$, CI $[0.606, 0.823]$). Grouped by craton, the signal is consistent: Yilgarn $\langle r \rangle = 0.65$, Abitibi 0.77, West Africa 0.68, with a two-sample KS test between Yilgarn and Abitibi finding no difference ($p = 0.83$). The repulsion is therefore a cross-cratonic feature of orogenic gold, not a peculiarity of one belt. Because no magmatism is involved, this repulsion cannot be inherited from magmatic-intrusion timing—it points to an intrinsic charge-release memory in the ore system itself (here, repeated fluid focusing along reactivated shear zones).

3.3 Grand unification across five systems

Combining with mantle plumes (Chen, 2026c), five physically unrelated systems—core–mantle boundary plumes, subduction-arc porphyry Cu, intracontinental granite W–Sn, collisional porphyry Cu, and metamorphic orogenic gold—all show single-source repulsion that collapses to Poisson on mixing (Figure 2). The source-separation effect ($\langle r \rangle_{\text{single}} - \langle r \rangle_{\text{mixed}}$) ranges $+0.13$ to $+0.32$. This consistency across radically different mechanisms, depths, and timescales is the strongest evidence for an underlying principle.

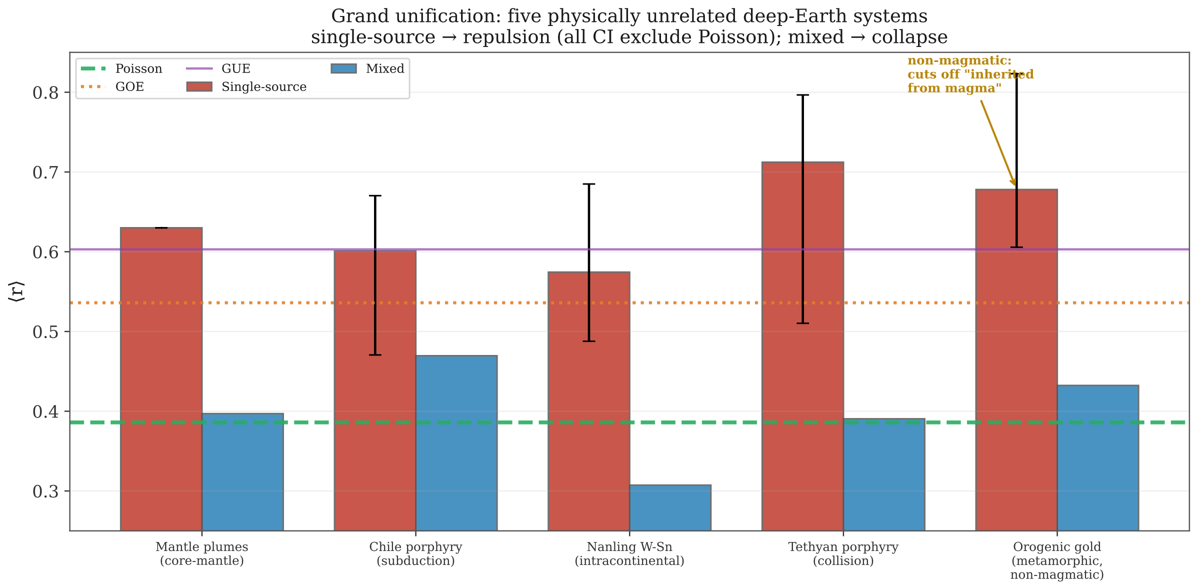

Figure 2: Five physically unrelated deep-Earth systems. Single-source configurations (red, with bootstrap CIs) all exclude Poisson and reach the GOE–GUE range; mixed configurations (blue) collapse. The non-magmatic orogenic-gold case (annotated) cuts off the “inherited from magma” explanation.

4 Discussion

4.1 A unifying principle

Across five systems a single principle organizes the results: *GOE-class level repulsion emerges wherever a single long-memory process is isolated, and Poisson emerges wherever many independent sources superpose*. The charge-and-release mechanism for ore systems is the metallogenic instance: a pulse exhausts a local resource that requires time to replenish, forbidding short recurrence intervals.

4.2 Progress on intrinsic versus extrinsic

The non-magmatic control advances the central question. Were the repulsion merely inherited from magmatic cyclicity, orogenic gold—lacking magmatism—should not show it; yet it shows among the strongest repulsion of the four provinces. This favours an intrinsic interpretation: ore systems possess their own charge-release memory, independent of magmatic forcing. The argument is indirect (a mechanism contrast, not a per-district pairing), and the definitive test remains comparing, within a single district, the RMT spectra of ore-mineral versus magmatic-mineral ages—which requires datasets dense in both, currently unavailable. But the non-magmatic result removes the simplest extrinsic explanation.

4.3 Limitations

Sample sizes are modest (21–40 pooled intervals per province); dating methods are heterogeneous; economic-selection bias over-represents giant deposits; and orogenic gold operates on Gyr-aged cratons with Myr-scale inter-pulse intervals, a different timescale from the Myr-aged porphyry systems with 10^5–10^6 yr pulses. GOE-versus-GUE is unresolved at present sample sizes. A systematic database-level follow-up is the natural next step.

5 Conclusions

Single ore-forming systems in four metallogenic provinces—spanning subduction, intracontinental, collisional, and metamorphic (non-magmatic) settings—show GOE-class level repulsion in their multi-stage timing, collapsing to Poisson when mixed. The non-magmatic orogenic-gold control shows that the repulsion is not merely inherited from magmatic cyclicity, supporting an intrinsic charge-and-release memory. Together with mantle plumes, the result unifies five physically unrelated deep-Earth systems under one principle: single long-memory sources repel; superposed sources randomize. The definitive intrinsic-versus-extrinsic test awaits per-district datasets dense in both ore and magmatic ages.

Code and Data Availability

All code, compiled datasets, and figures: https://github.com/Ruqing1963/metallogeny-rmt.

Companion studies: https://zenodo.org/records/20766310 ; https://zenodo.org/records/20768130 ; https://zenodo.org/records/20768420.

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